

# Gesture Based Controller For Film Drone

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**Abstract**—This paper proposes a method that combines geometric-based body pose and hand gesture recognition to develop a human-drone interaction (HDI) based controller for a DJI Tello drone that can be operated in real time with only a laptop. The proposed method combines MediaPipe-based landmark extraction with geometric hand gesture classification to give drone instructions. MediaPipe-based body pose tracking was implemented alongside this gesture classification, with the extracted positional data enabling proportional drone control via the RC control function provided by the DJI Tello software development kit. This proposed method provided gesture control of the drone's elevation, horizontal position, and rotation. It also enabled the drone to follow the operator, either continuously facing them or with its rotation locked. The performance of this method was evaluated across 10 trials of a predefined routine, tracking how many times gestures were misidentified or the drone exhibited behaviour not assigned to the current instruction. This test showed that the proposed method correctly identified hand gestures with 100% accuracy, performing similarly to machine learning-based approaches from prior research that achieved 90% to 99.7% accuracy; however, these approaches were evaluated on larger sample sizes, making their performance numbers more reliable. This testing also showed that the proposed method exhibited correct behaviour with 100% accuracy in 5 of the 6 drone control instructions, with follow mode performing correctly with 80% accuracy, making it less reliable than prior research that employed a similar approach to follow the drone operator.

**Index Terms**—drone, gesture control, DJI Tello, film, MediaPipe, pose tracking, hand tracking, human-drone-interaction

## I. INTRODUCTION

Drones have been among the most transformative technologies in media in recent years [1], becoming an important tool for cinematographers [2] and utilised in a wide range of visual media applications, such as journalism, videography, and advertising [3]. Drones have opened new possibilities for filmmakers, allowing them to capture shots that were previously expensive and difficult to achieve, offering a more affordable alternative and making aerial cinematography more accessible to both big-budget productions and independent filmmakers [1]. However, a key limitation of drone cinematography is the need for a crew member to be dedicated solely to piloting the drone, leaving them unavailable for other tasks [3]. The drone operator must also possess the skills to fly the drone, including an in-depth understanding of its dynamics and controls [3]. To combat this, drones today are moving towards the inclusion of AI features and Human-Drone Interaction (HDI) approaches, such as gesture control [3]. The

development of these features has the potential to make drone control more intuitive, as improvements in drone technology continue to drive innovation, further shaping the way stories are told on screen [1]. This paper proposes a method that combines geometric body pose and hand gesture recognition to develop an HDI-based controller for a DJI Tello drone, mapping these gestures to movement instructions for the drone that cover common drone applications in filmmaking.

## II. BACKGROUND

### A. Existing Approaches

A common approach to HDI is training a machine learning model to recognise predefined gestures from a camera input, with each gesture mapped to a drone command [4], [5], [6], [7], [8], [9]. One method is to use MediaPipe to extract landmarks from an image, then train a gesture recognition model using these landmarks, mapping each gesture to a command for the drone [4], [6], [7], [9]. This approach achieves high accuracies between 95.7% and 99.7% [6], [7], [9]. However, the performance of this approach is limited to its training data, requiring curated datasets with large numbers of images per gesture class. Models trained on fewer images per gesture class can see a drop in accuracy, as shown in [4], which reported a 90% to 95% accuracy. However, this approach is vulnerable to performance degradation when exposed to images not represented in the model's training data, such as varying lighting conditions [4], [7] or partial occlusion [7], leading to potential false positives [4] that can result in unpredictable drone behaviour.

Another approach is to train a Convolutional Neural Network (CNN) to recognise gestures in video [5], [8]. CNNs require a balance between accuracy and computational performance. T. Begum et al. recorded an accuracy of 92.5% when evaluated on a computer [5], while Khoza et al. developed a model achieving an accuracy of 80%, but their model was limited by being designed for low computational power devices, resulting in objects being recognised incorrectly due to poor video quality [8].

Machine learning-based gesture recognition algorithms achieve high accuracies but are limited by the image dataset they are trained on, with varying conditions in the live video feed potentially causing gestures to be incorrectly identified [4]. This approach also increases the resources required to implement new gestures, as the model must be retrained,

which requires large numbers of training samples to create the new class.

Another method of implementing HDI is the use of a geometric approach, using key points on the operator's body to derive drone controls [10], [11]. N. Gio et al. used a Microsoft Kinect to capture the operator's body posture, calculating angular values of the operator's body to recognise their movements. The magnitude of these relative angles was used to control the drone's speed for each movement [10]. M. Bhat et al. used 2D coordinates of 18 key points to track the operator's pose, using a model they implemented in PyTorch trained on COCO and CMU panoptic datasets. Using this pose estimation, they mapped pose actions to drone commands, using the nose landmark to control the drone's thrust and yaw while also keeping the operator's face centred in the frame.

Alongside general flight control, both [11] and [10] developed methods to give more complex commands. M. Bhat et al. implemented a method to track the user's wrist so they can draw letters corresponding to different drone instructions, such as drawing a 'P' to take a panoramic shot [11], while N. Gio et al. developed a graphical interface with an interactive menu that allowed the user to select a command by moving their right arm like a clock hand to the segment containing the desired drone instruction [10].

This approach to drone control demonstrated potential, with first-time users finding it much easier to control compared to a smartphone controller [11]. Users with prior drone piloting experience were faster with a standard controller, but this gap narrowed as they became more familiar with gesture control [10]. Body gesture control was also found to be easy for inexperienced operators to learn, requiring only a brief explanation and a few seconds of practice to mostly avoid further clarification, whereas the joystick controller required multiple clarifications of the controls [10]. The difference recorded in [10] between the two control methods would likely have been smaller, as the results were limited by a significant 1.3-second latency between commands, which lengthened the experimental task for the gesture controller.

Kimata et al. presented an approach that combined MediaPipe geometric-based hand gesture recognition with Haar Cascade face detection, utilising the user's face position to keep the drone at a safe distance and centre the camera on the user. This system was tested with a Tello drone, detecting gestures with a 94.27% accuracy, but was limited by the facial recognition as it generated a significant number of false positives, with clothing patterns or the desk in frame being mistaken for a face due to the Haar cascade classifier's reliance on contrast patterns to identify faces. This resulted in facial detection achieving 34.04% accuracy, along with additional issues arising from gestures that occluded the operator's face. This low facial recognition accuracy led to a low distance maintenance accuracy of 21.71%, as facial recognition was a prerequisite for determining the distance between the drone and the operator [12].

Another method of implementing HDI control was through a wearable controller with specialised hardware that communicated instructions to the drone [13], [14].

M. Yoo et al. implemented this approach using a thumb-mounted Inertial Measurement Unit (IMU) to control the drone's continuous inputs, such as movement direction, while a hand gesture recognition system provided drone operation commands, such as takeoff and landing. This setup achieved 98.3% gesture recognition accuracy and a 0.089ms processing time, with gesture-controlled lap times in a simulation environment averaging 19 seconds less than joystick-controlled lap times. Because each user's hand position and gesture characteristics vary, an alignment process had to be performed before use, requiring the user to remain stationary for 5 seconds at the start of operation [13].

K. Konstantoudakis et al. had a different approach, using a Microsoft HoloLens 2 augmented reality headset to provide the user with a visual overlay of the drone's position and flight data. The headset's cameras, along with the Mixed Reality Toolkit (MRTK) Unity3D plugin, were used to provide palm-based control for the drone. This aimed to provide users with greater context for how the drone is moving. Initial feedback from simulated drone flights suggested that gesture control was easy to learn and comfortable to use for inexperienced users. However, under certain outdoor conditions, such as very bright sunlight or high temperatures, the performance of the palm tracking dropped, and high temperatures caused the HoloLens to shut down after a few minutes due to its reliance on passive cooling. The manual calibration for the drone-tracking displayed on the headset also often proved too complicated for users [14].

Wearable controllers and hardware offer a good approach to HDI but are limited by their requirement to be tethered to other devices, either by physical means [13], which restricts freedom of movement, or via a wireless connection [14], which can introduce latency if commands are communicated through multiple devices. This also affects their use in media production, as the user must wear visible hardware, which means they cannot be in frame at the same time, making this approach unsuitable for smaller projects where the user is often filming themselves.

Several of the surveyed existing approaches were not tested on a physical drone but were evaluated only on computers and simulators [5], [9], [13], [14], [15]. This limits confidence in the reported performance figures under real-world operating conditions, where connection latency, physical drone behaviour, and environmental factors would significantly influence performance.

## *B. Gap Analysis*

Although the reviewed approaches demonstrate the viability of gesture-based HDI, several limitations remain, motivating the work presented in this paper.

A large number of these existing approaches rely on trained classifiers to recognise gestures, which require large datasets and need retraining if a new gesture is added [4], [5], [6], [7], [8], [9]. While this approach offers high accuracy, it limits flexibility by increasing the effort required to add new gestures and is vulnerable to conditions not represented in the training data. Furthermore, as this approach is based on discrete classification, it prevents proportional speed control for the drone, where the response is scaled based on the operator's position or actions, thereby limiting the ability to implement features such as a follow mode.

Existing methods in these papers predominantly use either hand or body gestures to control a drone, with [4], [6], [8], [9] considering only hand gestures and [10], [11] considering only body gestures. A combination of both potentially offers a more user-friendly operation experience and separates drone instructions from movement control.

A subset of these existing approaches was evaluated only as gesture recognisers or within drone simulators [5], [13], [14], [15], resulting in reported results that did not address the complications attributable to connection latency, range limitations, and environmental conditions.

A few approaches require additional hardware that has to be worn by the operator to control the drone [13], [14]. This increases the cost of the approaches while making them unsuitable for filming when the operator is also expected to be in frame, such as solo productions.

The following sections present a geometric-based HDI method developed to address these limitations.

### III. PROPOSED METHOD

#### A. Hardware Setup

This system uses the DJI Tello drone's front-facing camera, streaming the video feed over Wi-Fi to a laptop, where each frame is processed and converted into a set of instructions for the drone using the Tello SDK. These instructions are then sent back to the drone over Wi-Fi. To prevent the video feed streaming, hand and body pose processing, and drone instruction calculations from interrupting each other, threading was used to process them individually on separate CPU cores, controlling how frequently each is executed. The video feed from the Tello camera was limited to 15 fps, as this was enough for the gesture recognition to operate smoothly without processing an excessive amount of frames.

#### B. Pose and Hand Tracking

To implement a geometric-based gesture controller, MediaPipe was employed, providing the landmark data for gesture recognition and drone control. Hand tracking was configured with a minimum detection confidence of 0.3; the lower detection threshold was necessary to reliably identify the operator's hands at greater distances. Pose tracking was configured with a minimum detection confidence of 0.4 because the operator's body occupied a significant portion of the screen, making detection easier. A minimum tracking confidence of 0.5 was used for both hand and pose tracking,

as it was sufficient to maintain landmark tracking once they were initially detected.

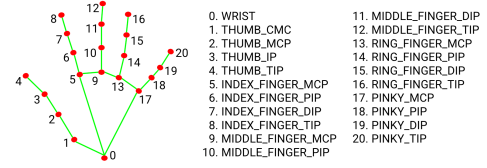


Fig. 1: MediaPipe hand landmarks.

Hand gesture recognition was used to instruct the drone to switch to the appropriate control mode (Fig. 2). To implement this, a finger class was created, with each finger instance containing the finger's name, the hand it was on, and a function to check whether the finger was extended. This function checked if the length from the fingertip landmark to the wrist landmark was greater than 1.5 times the length of the finger knuckle landmark to the wrist landmark (Fig. 1). Gestures were constructed by specifying which fingers were extended and which were not. Checking whether the thumb was raised was also determined using the same approach, with the y-position of the thumb tip landmark relative to the wrist landmark distinguishing between a thumbs-up and a thumbs-down gesture.



Fig. 2: Hand gestures and corresponding instructions.

Body pose tracking was used to control the drone's movement and indicate that the operator was ready to send a new instruction, with MediaPipe world landmarks (Fig. 3) used to calculate several metrics. First, the distance from the shoulder and wrist landmarks to the hip landmark was used to determine whether the user's hand was raised, by checking whether the wrist-to-hip length was greater than 1.5 times the shoulder-to-hip length. The other metrics calculated from these world landmarks are the distance between the operator and the drone, the midpoint between the operator's shoulders, body yaw, and the angle between the operator's shoulder and wrist. These remaining metrics are detailed in Section III-D, along with the associated drone control modes that utilise them.

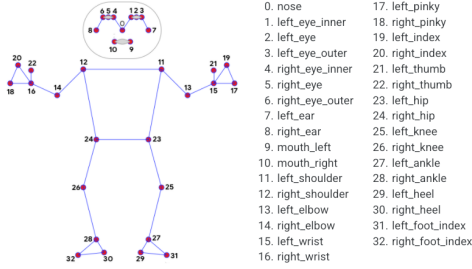


Fig. 3: MediaPipe pose landmarks.

### C. Drone State Logic

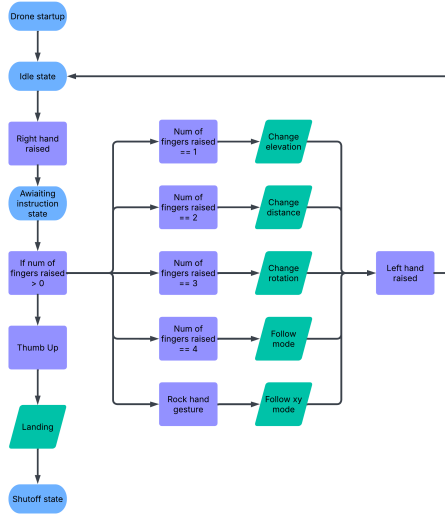


Fig. 4: Drone state logic flow chart.

Fig. 4 illustrates the state machine used to control the drone, preventing multiple instructions from executing simultaneously and enabling robust control system operation. When the system is initialised, the drone enters the start-up state, connects to the user's laptop, enables video streaming, and then takes off, rising to a height of 150cm before transitioning to the idle state. In the idle state, pose tracking is enabled, checking whether the operator has raised their right hand, and moving the drone into the awaiting instruction state if this

right hand is detected. In this state, pose tracking is disabled, and hand tracking is enabled, with the program checking how many fingers are raised on the operator's right hand or whether another gesture is displayed (Fig. 2). The operator must hold this gesture for 3 seconds for the signal to be considered accepted. This was implemented to reduce the risk of false positives. If a thumbs-up gesture is accepted, the drone will transition to the landing state and land before disconnecting. If a valid gesture is raised for the three-second period, the drone will transition to the drone control state associated with that gesture (Fig 2). Upon receiving the instruction to switch to the drone control state, pose tracking will be enabled to track the relevant body landmarks required to control the drone in that state. In each instruction state, the drone will follow the commands for that state until the operator's left arm is raised, at which point the drone's state will transition back to idle. This approach reduces processing resources by only allowing either hand or pose tracking at a time.

### D. Drone Control

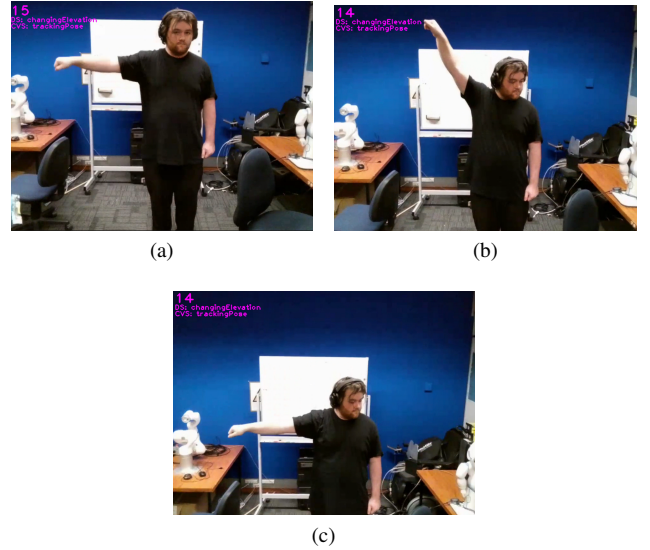


Fig. 5: Changing elevation with body gestures.

Five drone movement control sets were used to cover common drone applications in filmmaking: changing elevation, changing distance, changing rotation, follow mode, and follow xy mode, along with a landing mode, each mapped to the relevant gesture (Fig. 2)

$$\begin{aligned} dx &= \text{rightShoulder } x \text{ pos} - \text{rightWrist } x \text{ pos} \\ dy &= \text{rightShoulder } y \text{ pos} - \text{rightWrist } y \text{ pos} \\ \text{angle} &= \text{atan2}(dx, dy) \end{aligned} \quad (1)$$

In the elevation control mode (Fig 5.), the angle (1) between the operator's right shoulder and wrist is used to control the drone's elevation: the drone moves up when the angle is less

than 70 degrees and down when it is greater than 110 degrees. This command is sent via the RC control function in the Tello SDK, with the rate of ascent or descent proportional to the error between the drone's current height, as measured by the onboard time-of-flight sensor, and either the maximum or minimum height constraint set by the operator.

$$distance = \frac{known\ width * focal\ length}{pixel\ distance} \quad (2)$$

Distance is controlled by using the angle between the operator's right shoulder and wrist, along with the estimated distance between the drone and the operator (2), using the same approach as in elevation control mode to move the drone closer or further away. This distance is set to a maximum and minimum range to prevent the drone from colliding with the operator or travelling too far out where the operator can no longer be detected.

$$\begin{aligned} avg\ x &= \frac{left\ shoulder\ x + right\ shoulder\ x}{2} \\ pixel\ avg\ x &= avg\ x * image\ width \\ pixel\ offset &= avg\ x - frame\ center\ x \end{aligned} \quad (3)$$

To change the drone's rotation, the pixel offset from the midpoint of the operator's shoulder landmarks to the frame's centre is calculated (3). When the offset is greater than 30px from the centre of the camera, a proportional controller uses it to calculate a yaw speed, rotating the drone to centre the operator in the frame.

In follow mode, the drone maintains a fixed position relative to the operator, using the estimated distance between the drone and the operator (2) along with the rotation control set. Follow mode continuously adjusts the drone's heading, keeping the operator centred in the frame within a pixel threshold. If the operator is centred within this threshold, the drone will adjust its distance from the operator to maintain the follow distance. The drone first prioritises centring the operator before adjusting its distance, ensuring it will always move toward or away from the operator along the operator's shoulder axis.

Follow xy mode maintains a fixed position relative to the operator, using the estimated distance between the drone and the operator (2). However, instead of continuously adjusting the drone's heading, it calculates the pixel difference between the midpoint of the operator's shoulders and the centre of the frame (3). This distance is used in a proportional controller to determine the drone's sideways speed, allowing it to move left/right and forward/back following the operator, keeping them centred in the frame and at a fixed distance.

To ensure the drone's safe operation, speed checks are performed in all control states, limiting the drone's maximum

speed to 30 cm/s. In addition to the drone's limited speed, user-defined maximum and minimum distance and elevation constraints prevent the drone from colliding with obstacles or the operator. During drone operation, an empty RC command is sent to the drone as a keep-alive signal every 10 seconds. This was required to override the drone's behaviour and prevent it from landing during states where no movement commands were being sent, as the Tello SDK causes the drone to land if no command is sent within a 15-second period. This interaction means that if the drone disconnects, it will land itself if it is unable to reconnect and receive the keep-alive signal.

## IV. RESULTS

### A. Testing method

To evaluate the effectiveness of this approach to drone control, a test was undertaken in which a predefined routine was performed 10 times in a consistent environment with the same operator, analysing the number of times the proposed method misinterpreted instructions and whether the drone or control software exhibited unwanted behaviour. This test included all six drone control modes defined in Section III, performed in a fixed sequence, with each trial instructing the drone to change elevation (up then down), change distance (back then forward), rotate (90° left then back), follow x,y, follow the operator, and finally land.

### B. Hardware and software specifications

This test was performed on an HP Spectre x360 laptop running Windows 11, with an Intel(R) Core(TM) i7-1065G7 CPU @ 1.30 GHz and 16 GB of RAM. The software was compiled in Visual Studio Code using Python 3.12 (64-bit), OpenCV-Python version 4.13.0.92, and Mediapipe-Numpy2 version 0.10.21. The camera used for hand and pose tracking was the DJI Tello drone camera, with HD 720p video at 30 fps.

### C. Results

The instruction recognition results are shown in Table I, with the gestures for all six drone control modes being recognised with 100% accuracy across all 10 trials. The drone's behaviour during the test routines followed this trend, performing the correct behaviour with 100% accuracy in five of the six control modes; however, the accuracy in the follow mode was only 80%, as shown in Table II. The incorrect behaviour was observed when the operator left and subsequently re-entered the frame, causing the drone to be sent back to the awaiting instruction state. This occurred as the function that checks whether the operator's hand is raised is not required to remain active for a minimum amount of time, so when the operator re-enters the frame, only their wrist landmark may be visible, leading the system to incorrectly interpret this as both hands being raised and reverting the drone through the idle state to awaiting instructions. During testing, the drone's video feed was limited to 15 fps to maintain the program's stability. This 15 fps was consistently achieved throughout all trials, with only minor variations observed.

TABLE I: Instruction Recognition Results

|                  | Instruction             |                        |               |                  |               |             |
|------------------|-------------------------|------------------------|---------------|------------------|---------------|-------------|
|                  | <i>Change Elevation</i> | <i>Change Distance</i> | <i>Rotate</i> | <i>Track x,y</i> | <i>Follow</i> | <i>Land</i> |
| <b>Correct</b>   | 10                      | 10                     | 10            | 10               | 10            | 10          |
| <b>Incorrect</b> | 0                       | 0                      | 0             | 0                | 0             | 0           |
| <b>Accuracy</b>  | 100%                    | 100%                   | 100%          | 100%             | 100%          | 100%        |

TABLE II: Instruction Behaviour Results

|                  | Instruction             |                        |               |                  |               |             |
|------------------|-------------------------|------------------------|---------------|------------------|---------------|-------------|
|                  | <i>Change Elevation</i> | <i>Change Distance</i> | <i>Rotate</i> | <i>Track x,y</i> | <i>Follow</i> | <i>Land</i> |
| <b>Correct</b>   | 10                      | 10                     | 10            | 10               | 8             | 10          |
| <b>Incorrect</b> | 0                       | 0                      | 0             | 0                | 2             | 0           |
| <b>Accuracy</b>  | 100%                    | 100%                   | 100%          | 100%             | 80%           | 100%        |

#### D. Limitations

While the proposed approach demonstrated high accuracy in the controlled conditions it was evaluated under, several limitations were identified. A key limitation identified was the performance of the hand tracking when the drone was approximately 1.5 metres or greater from the operator, resulting in issues occurring when trying to initially detect the operator's hands, with the operator having to display an open hand closer to the drone before switching to holding up the gesture to switch into the drone control mode they desired. A further limitation was that the system could only reliably operate with a single person in frame at a time, as having multiple people in frame would cause the pose tracker to switch between them based on confidence levels, resulting in incorrect operator identification. Another limitation was the sample size of the test used to evaluate the proposed system, which included only one operator performing ten routines under consistent conditions. As such, the performance of the proposed system is not generalisable and could degrade when under different operating conditions or when controlled by a different operator. A final limitation was the occasional disconnection of the DJI Tello drone observed throughout the system's development. While the drone automatically reconnected in most instances, manual reconnection was required several times.

#### V. CONCLUSION

This proposed method aimed to provide a human drone interaction approach as an alternative to joystick control, allowing an operator to control all aspects of the drone's movement with hand and pose geometric gestures. This was achieved by implementing MediaPipe pose and hand tracking, along with the DJI Tello Software Development Kit (SDK). The proposed method achieved 100% gesture recognition accuracy for drone instructions, with the drone performing the correct behaviour with 100% accuracy in five of the six control modes and 80% accuracy in the follow mode.

##### A. Comparison to prior research

The machine learning-based approaches reviewed in Section II achieved gesture recognition accuracies of 90% to 99.7% using a trained MediaPipe classifier approach [4], [6], [7], [9].

The CNN-based approaches achieved accuracies of 92.5% [5] and 80% [8]; however, the results in [8] may be attributed to the fact that it was designed for use on platforms with minimal processing power. The method proposed in this paper achieved 100% accuracy in gesture recognition. However, this was only on a sample set of 10 trials under consistent conditions, making direct comparison with prior work difficult, as the reviewed systems were evaluated across more varied conditions and larger datasets.

Both [7] and [12] implemented a form of distance control between the drone and the operator. The approach outlined in [12] experienced frequent failures, achieving only 21.71%, as it relied on distance estimation to the operator's face, detected using the Haar Cascade face detector, which generated a significant number of false positives, achieving only 34.04% on its own. The system presented in [7] successfully achieved human following functionality, using 2D and 3D coordinates of the body landmarks obtained through MediaPipe to determine the relative position of the operator from the drone, using a proportional-integral (PI) controller to control the distance, angle and height of the drone, sending velocity commands to the Tello Drone via the RC control function from the Tello SDK. The method proposed in Section III used a similar approach in the design of both normal and xy follow modes, using the distance and pixel offset estimated from the operator's 3D shoulder landmarks to control the drone via proportional values; however, both drone modes omitted height control. This approach achieved 80% correct behavioural accuracy in follow mode, and 100% in xy follow mode, with the failures in follow mode occurring only in an edge case where the operator left and re-entered the frame.

The evaluation of the proposed method in this paper was undertaken using a DJI Tello drone, allowing the results to reflect how this approach performs under connection latency, range limitations, and environmental conditions encountered in drone operation. This reflects real-world operating conditions more closely compared to methods that were only tested on a computer or in a simulated environment [5], [9], [13], [14], [15].

##### B. Future Research

Future development of this system could focus on implementing a method to isolate the operator from other people in the frame. This would overcome the limitation that the drone only operates correctly when one person is in the camera's view, allowing the drone to be used when multiple people need to be filmed or when out in public. Another area for future improvement is the method used to switch between drone instructions, replacing the current approach of using finger gestures with a system based on pose tracking, such as those in [10], [11], or a semaphore-inspired method. This would overcome the drone's current distance limit, allowing it to be controlled over a greater range and making it more suitable for filming applications.

## ACKNOWLEDGEMENTS

AI usage: Claude and Grammarly were used for punctuation, clarity checking and spell checking throughout this paper. Claude was also involved in the development of drone control software, providing further details on the DJI Tello SDK and for bug testing.

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